

Utilizing Prosodic Analysis for Music Emotion Recognition

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1 Introduction

Emotions have long been centered at the core of music experience and creation, often cited as inspiration for a creative work that later shapes the mood of the listener [18]. The ability to computationally classify the emotions that listeners experience from particular pieces of music has potential to benefit fields such as music therapy, recommendation systems, and the development of person-centered healthcare [1]. However, this task is far easier for humans than computers, motivating the field of music emotion recognition, or MER.

Despite steady progress in MER, current state-of-the-art (SOTA) systems remain well below human-level performance, particularly when handling vocal music. Existing approaches typically treat lyrics and audio as separate modalities that are only combined at the sentence level, neglecting the prosodic detail of how individual words are sung. This raises the central research question of this study: *can integrating prosodic analysis into a multimodal MER framework improve music emotion prediction on tasks at varying levels of processing granularity?* Building on prior evidence that prosody is a strong cross-cultural carrier of emotion in speech, this study hypothesizes that integrating prosodic features alongside lyrical and acoustic features within a cross-modal architecture will yield measurable gains in regression metrics at the word granularity.

2 Literature Review

Accurate MER systems depend upon the contribution of multiple components: a representation of emotion that captures complexity, a set of features extracted from music, and an architecture that sufficiently links the features to the output.

2.1 Emotional Model

The task of computationally recognizing emotions relies upon the method of reliably representing representing the complexity of human emotions within a data structure. To develop accurate methods of MER, an understanding of the various models of emotions is necessary. Most emotion models take on two distinct types of representation: continuous, or discrete. Those of the former type often represent a particular emotion as a vector in a multidimensional space, where axes are determined based on theoret-

ical factors in emotion expression. For example, Russell’s circumplex model uses valence¹ and arousal² as two axes, while Thayer’s Model bases dimensions on energy-stress and calm-tired ranges [14, 17]. Discrete models, on the other hand, involve the definition of a set of emotions and characteristics. In the context of music, two discrete models are popular: the Geneva Emotional Music Scale Model, and Hevner’s Emotional model, both of which rely on descriptions of the emotional response to music [13, 24]. Though the aforementioned discrete models have merit in their application to musical terms, surveys have shown that most MER researchers choose Russell’s Model due to dataset availability [10].

2.2 Features

Equally important to the way the output of computer models for MER is represented are the inputs that are given to the computer model, or the features of the MER model. In this aspect, MER models vary much more with the type and level of features utilized. Historically, MER features have been organized into a hierarchy of levels based on the complexity and degree of processing required to extract [3]. Low-level features usually are focused on the basic acoustic properties of music, extracted from short audio segments without context of the greater music piece. Included in this classification are two main categories of features: spectral features, and temporal features. Spectral features aim to capture timbre and tonal properties of music by employing methods such as mel-frequency cepstrum coefficients³, zero-crossing⁴, spectral centroid⁵, spread⁶, roll-off⁷, and others [6, 7, 11]. These features are commonly

¹Valence describes whether an emotion is positive (pleasant) or negative (unpleasant).

²Arousal describes the energy or activation level of an emotion, ranging from calm to excited.

³Mel-frequency cepstrum coefficients (MFCCs) are a compact numerical summary of an audio segment’s frequency content, designed to mirror how humans perceive pitch.

⁴The zero-crossing rate is the number of times an audio waveform changes sign per unit time, roughly indicating how percussive a sound is.

⁵The spectral centroid is the “center of mass” of a sound’s frequency spectrum, or a measure of where most of the signal power in a sound lies. It is often perceived as a measure of brightness.

⁶Spectral spread measures how widely a sound’s energy is distributed around the spectral centroid.

⁷Spectral roll-off is the frequency below which a specified percentage (commonly 85%) of the total spectral energy lies.

extracted using spectrograms, or analysis of the frequencies of audio over time, and have dominated early MER research due to the efficiency and direct correlation to audio. Temporal features, on the other hand, capture the rhythm and tempo of a song, along with other time-related aspects of the track. Temporal features differ in comparison to spectral features, as they typically focus on how aspects evolve over durations of time. Though this technique has not been utilized as much as spectral features in the context of MER research, analyses of time-variation features, such as the consistency of rhythmic patterns, demonstrate a correlation between evolving temporal features throughout time and the respective emotions elicited from music [22].

In MER research, most models rarely utilize features above the low-level section, especially due to the complexity that arises when processing mid- and high-level features. Thus, there has been limited use of mid-level features such as pitch contour⁸, melodic patterns, and prosodic variations, or the combination of rhythm, pitch, stress, and intonation to create a sense of musical quality [10], despite the success in other related music information retrieval tasks, like cover-song detection [15]. The gap in research is particularly notable given that mid-level features seem to bridge the divide between the low-level features of acoustic description and high-level semantic understanding, thus potentially being able to assist in a more interpretable and emotionally relevant representation of musical content.

Historically, lower-level descriptions were hand-crafted to maximize information extraction, resulting in greater levels of accuracy. However, recent usage of learned embeddings, such as the VGGish convolutional network [5] pretrained on AudioSet log-mel spectrograms producing an 128-dimensional embedding per audio segment, has begun more comprehensively represent audio streams.

2.3 Techniques

The evolution of MER systems reflects the increasing sentiment that emotional content in music cannot simply be encapsulated through audio analysis alone. Early approaches analyzed the audio features of timbre, tempo, and spectral characteristics separately from the lyrical content processed through natural language processing (NLP) techniques. However, this separation, albeit much simpler, failed to account for the relationship and interconnection between what is sung and how it is sung. The semantic content of the lyrics does influence the emotional perception, but the variation with which words are delivered in vocal performance adds many layers of nuance that separation could not capture. As in speech, the way words are said matters as much as which words that are said. Recent work has begun to address this shortcoming via the incorporation of lyrical,

auditory, and even visual data into multimodal frameworks. However, many of these approaches still exhibit the same limited interconnection seen in prior MER systems [21, 22]. This is mainly because each modality is typically processed through individual feature extraction pipelines and neural network pathways, resulting in representations that are only concatenated at later stages of the model. Late concatenation typically prohibits models from learning how details such as a sudden vocal intensity spike interacts with a lyrically significant word, primarily because each modality contributes its own prediction, causing adjustments in either modality to be lost in the final prediction head.

2.4 Current SOTA

Current state-of-the-art models have varying degrees of cross-modal integration, or the interfacing between each type of feature through the network pathways. The COSMIC framework by Yang et al. introduced emotion-long-short-term-memory (emotion-LSTM)⁹ cells that allow for cross-interaction between lyric and melody representations of music tracks, yet also included an implementation of a structural analysis approach that processes the verse and chorus sections separately [22]. The structural awareness included recognizes the idea that different sections of a song can serve different functions, with sections building on each other to develop narrative or thematic content. The emotion-LSTM cells utilized in the COSMIC framework enable the bidirectional information flow between the lyrical and acoustic features typically seen in modern natural language processing architectures. This allows the model to learn how the different modalities are developed throughout the song to produce emotional impact. Expanding on this, the MMD-MII model proposed by Wang et al. adds on a multilayered analysis approach with a dedicated cross-processing module, representing another step in the direction of more multimodal integration [21]. The MMD-MII framework similarly processes the chorus separately from the verses, again acknowledging the intricacies in music structure. The dedicated cross-processing module allows greater interaction between the lyrical content and musical content of the audio input for greater cross-modality. However, these models are both still limited by the processing of vocal content at the coarse unit of a phrase, with each lyrical line treated as a unified semantic and acoustic unit without examining whether the variation within the unit carries additional emotional signal. An analysis at the word-level could potentially be revealing regarding further data hidden between overall phrases.

This processing and analysis at the sentence level instead of a finer word-level overlooks a crucial part of emotional expression in music: prosody. Prosody typically

⁸A pitch contour is the trajectory of perceived pitch over time within a melody.

⁹Long short-term memory (LSTM) is a type of recurrent neural network designed to retain information across long sequences, making it well-suited for ordered data such as text or audio.

encompasses the “suprasegmental”¹⁰ aspects of speech and singing, including pitch variation, intensity¹¹, duration, rhythm, stress patterns, and intonation [20]. These features operate at a finer granular level than the typical sentence-level analysis that occurs in the current state-of-the-art models, often varying significantly from word to word or syllable to syllable within a single phrase. In spoken language, prosody is shown to be a carrier of information about emotion, attitude, emphasis, and pragmatic meaning, as the difference between sarcastic and sincere statements lies in prosodic delivery of even individual words rather than word choice. In music, prosody can take on even greater importance, as singers commonly extend the constrained prosodic dictionary used for speech to include manipulations of pitch, timing, and dynamics. Skilled vocalists may stretch words for dramatic effect, change pitch for emotionally significant emphasis, or even apply vibrato to add a sense of warmth or intensity to sustained notes. These prosodic choices in how notes are sung represent intentional artistic decisions aimed at conveying certain emotions, yet are largely unexamined by current MER systems.

Unfortunately, despite prosody’s well-documented utility as a carrier of emotion, it has remained largely absent from MER systems. Past research has demonstrated that prosodic features are intrinsically linked with emotional responses, with this link holding even across cultural boundaries [2]. This consistency between cultures implies that the prosodic-emotional correlation may reflect more universal aspects of human communication, potentially giving greater value to an application of prosodic analysis to MER systems. MER systems including prosodic analysis could work across diverse musical traditions and languages. Moreover, musical aspects of language, especially the prosodic elements, have been shown to influence cognition similarly in both speech and music, indicating shared processing mechanisms between linguistic and musical understanding [12]. Given the success of emotion recognition utilizing word-level prosodic analysis in speech information retrieval [19], integrating the same mechanisms in MER may bring greater accuracy.

2.5 Gap

After an observation of the current state-of-the-art models, a gap clearly emerges, stemming from three different areas of lack of research. First, the concept of cross-modal interaction is quite novel, and thus has not been researched deeply yet. The limited research on cross-modality in state-of-the-art MER suggests that the technique has potential to be useful, but models like MMD-MII tend to process the cross-interaction at a unit of a lyrical line, leaving finer granularity untested. This study therefore evaluates the framework

at both a pair (sentence from lyrics paired with associated acoustic representation) granularity and a word granularity. Second, the use of prosodic analysis is extremely limited, as many studies focus on low-level features rather than mid-level features such as prosody. Third, studies have found that the use of more features is strongly linked to higher performance, further motivating the research of an extra layer of features [8]. These gaps elicit the following question: *can integrating prosodic analysis into a multimodal MER framework improve music emotion prediction on tasks at varying levels of processing granularity?*

This study aims to bridge all three gaps by introducing a cross-modal framework that integrates prosodic feature extraction with the state-of-the-art architecture of emotion-LSTM chains, thus adding a previously underused feature type and testing whether finer granularity improves overall performance.

3 Methodology

The goal of this study is to gauge the effect of including various prosodic features the impact of different granularity on performance. The methodology used is a modification of the methodology and architecture developed and enumerated in the MMD-MII paper by Wang et al. The overall methodology involves implementing the architecture described in the aforementioned paper, adding a prosodic input stream, and running an ablation over feature set and granularity to identify the impact of each change. The remainder of this section reviews and defends, in chronological order, the steps taken to train and evaluate this model.

3.1 Dataset

One dataset was chosen for use with the training of this model: Dataset for Emotion Analysis in Music (DEAM). This dataset contains 1,802 audio files, with a mixture of excerpts and full-length songs. DEAM is specifically made for use with MER tasks, as it contains annotations for both per-second valence and arousal values, as well as overall song-level valence and arousal. All emotion annotations are in the range $[-1, 1]$, where negative values indicate negative valence or low arousal and positive values indicate the converse. Of the 1,802 files, 801 contained lyrical vocals. Of these, 207 were filtered out for having no annotated unit inside the lyrical annotation window. For this study, per-second values were utilized rather than song-level values in order to enable finer unit-level alignment with the use of LSTM modules.

However, there is a discrepancy between the defined annotations provided in DEAM and the output of the MMD-MII architecture. While the dataset contains valence-arousal annotations, the MMD-MII architecture outputs as classification, despite claiming to have used the DEAM dataset in their training process. This means that there is an undocu-

¹⁰Suprasegmental features are properties of speech that span multiple individual sounds.

¹¹Intensity refers to the energy of a sound, perceived as loudness.

mented conversion between valence-arousal and the classes used in the paper that this research does not have access to. As a result, a slight modification to the architecture was made: the implemented architecture contains an output head of valence-arousal instead of classification. This should not impact the validity of results, as comparative and ablation testing will be able to identify the significance of a relative change in correlation of the trained model.

Each experiment uses a five-fold cross-validation at the song level. Within each fold, the dataset is split with 10% of tracks reserved for validation, 10% for testing, and the remaining 80% used for training. After potentially significant findings, the main pair-level comparison repeats the five-fold split under five different partition seeds, for a total of 25 paired folds for a verification of significance. The other versions of ablation use a single partition seed, for a total of five folds. Splitting is performed at the song level to prevent data leakage¹². The validation split is used only to monitor performance between epochs and to trigger early stopping in the case of insignificant improvement following 10 continuous epochs.

3.2 Preprocessing Pipeline

Prior to training, preprocessing was performed on all of the data, turning the raw track audio into data usable for machine learning. First, all track audio underwent vocal source separation¹³ via the same Spleeter model used in the MMD-MII paper [4]. This is necessary for adequate accuracy when extracting prosodic features from the vocal track, which requires the raw vocal audio.

Next, lyrics were automatically transcribed using OpenAI Whisper (a locally ran whisper-small model). Though initially developed for non-musical vocal transcription, Whisper has been found to achieve highly-accurate performance on automatic lyrics transcription when combined with source separation to isolate vocals [16]. Whisper also has the advantage of providing the word-level alignment needed for training without fine-tuning. After extracting word-level timestamps, the lyrics underwent minimal post-processing aimed to remove low confidence words¹⁴ and identify verse/chorus structure. In addition, tracks with no vocals were filtered out, as previously mentioned in the dataset section, to align with this study’s (and the MMD-MII) focus on music with vocals.

After transcription, the final stage in preprocessing was to align the per-second emotion labels from DEAM with the word-level timestamps from the Whisper transcription.

¹²Data leakage occurs when information from outside the training set influences the model. Splitting at the song level prevents excerpts from the same song from appearing in both training and test sets, which would inflate apparent performance.

¹³Source separation is the process of isolating individual components (such as vocals) from a mixed audio recording.

¹⁴Words flagged by the transcription model as having low certainty.

This was achieved through interpolation of the emotion labels to the word-level granularity according to the following equation.

For each word w_i with time boundaries $[t_{\text{start}}^i, t_{\text{end}}^i]$:

$$v_i = \frac{1}{t_{\text{end}}^i - t_{\text{start}}^i} \int_{t_{\text{start}}^i}^{t_{\text{end}}^i} V(t) dt \quad (1)$$

$$a_i = \frac{1}{t_{\text{end}}^i - t_{\text{start}}^i} \int_{t_{\text{start}}^i}^{t_{\text{end}}^i} A(t) dt \quad (2)$$

Where $V(t)$ and $A(t)$ are the continuous valence and arousal annotations respectively. As labels fall in per-second intervals, this was implemented as the weighted average of these labels during each word’s duration. The resulting word-level labels are resampled to the model’s axis when the pair granularity is used.

3.3 Feature extraction

After preprocessing, feature extraction was broken up into three types: acoustic features, prosodic features, and lyrical features. Acoustic features were extracted from the full audio track (vocals and accompaniment) using VGGish [5]. When run through the VGGish network, each audio segment is transformed into a 128-dimensional embedding, with each embedding overlapping a unit’s time span mean-pooled to give one 128-dimensional acoustic vector per unit.

Prosodic features were extracted using Parselmouth, a Python interface to Praat, which is the widely used tool in both phonetic and prosodic research. Additionally, librosa, a library for interfacing with audio, was used for extracting onset, chroma, key, and chord analysis. After continuous development and refinement, three versioned feature sets were implemented and ablated. prosody-v1 emerged initially as a static descriptor set. Soon, prosody-v2 followed as a new set of features capturing the dynamics and tonality track. Finally, prosody-v3 aimed to take the dynamics and tonality features of prosody-v2 and make them relative to the track. All are extracted per word and resampled to the model’s axis.

Table 1. Dimension count and contents of the versioned prosodic feature sets.

Variant	Dims	Source	Purpose
prosody-v1	25	vocals	static Praat descriptors
prosody-v2	46	vocals + mix	performance + tonality
prosody-v2 sidecar	29	track	key, tempo, rubato
prosody-v3	26	vocals + mix	track-relative performance

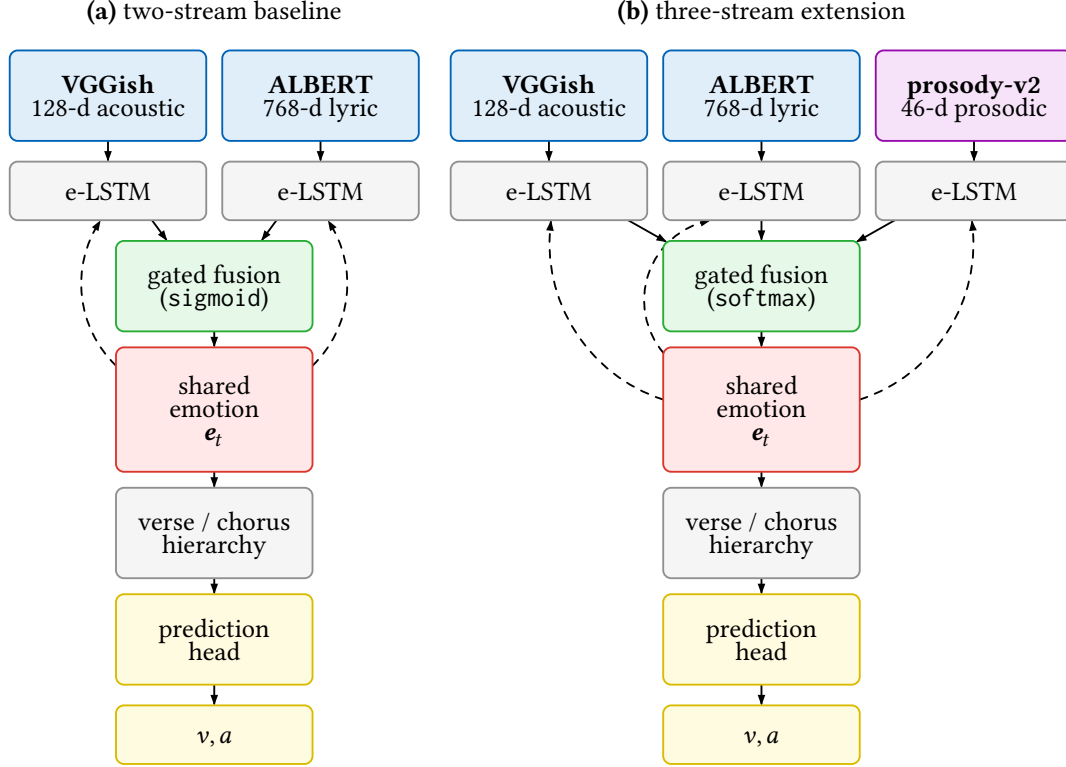


Figure 1. Architecture of the N-stream model. (a) The two-stream baseline combines a VGGish acoustic stream and an ALBERT lyric stream. (b) The three-stream extension adds a prosodic stream (prosody-v2 in this example). Each stream is encoded by an emotion-LSTM cell, and the candidate emotions are combined by a learned gate. Dashed arrows indicate the shared emotion vector e_t fed back into each cell.

Extraction used an F0¹⁵ floor of 75 Hz, F0 ceiling of 600 Hz, and time step of 10 ms. prosody-v1 comprises F0 mean, F0 standard deviation, F0 range, F0 slope, F0 velocity, voicing ratio¹⁶, jitter¹⁷ (local and absolute), shimmer¹⁸ (local and dB), HNR¹⁹ mean, HNR standard deviation, F1²⁰ mean, F1 standard deviation, F2 mean, F2 standard deviation, F3 mean, F3 standard deviation, F1 bandwidth, intensity mean, intensity standard deviation, intensity range, intensity slope, word duration, and syllable rate.

prosody-v2 adds performance features on the vocal stem (F0 relative to the track median in semitones, IQR, range, slope, rising/falling fraction, a six-bin pitch-interval histogram, interval mean/std, up-down and down-up transition rates, vibrato rate/extent, onset rate, beat residual,

and loudness variability) and tonality features on the mix (key-relative chroma, chord root, chord mode, chord match, tension, tension change, chord change).

prosody-v3 adds track-relative features: F0 median, IQR, range and slope; voice direction and mobility; F0 delta versus the previous word; speech rate; duration; gap to the previous word; beat residual; onset rate; relative loudness; loudness range and change magnitude; loudness delta; relative HNR; voiced fraction; jitter; chord change; chord match; tension and tension change; phrase-context F0 and loudness means; and position in the phrase.

For lyrical features, lyrics were encoded using ALBERT (albert-base-v2) [9], a lightweight transformer model in the BERT family. ALBERT was used with task fine-tuning, and the embeddings of a unit’s tokens were mean-pooled into a single 768-dimensional contextualized lyric vector per unit. This is consistent with the described implementation of the MMD-MII paper, thus maintaining a comparative basis with a state-of-the-art system.

Lastly, the prosodic features underwent z-score standardization before model input, with the following equation:

$$x_{\text{normalized}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}} + \epsilon} \quad (3)$$

¹⁵F0 (fundamental frequency) is the pitch of a sound.

¹⁶The voicing ratio is the fraction of analysis frames in which pitched speech is detected.

¹⁷Jitter is the cycle-to-cycle variation in F0, reflecting irregularity in vibration.

¹⁸Shimmer is the cycle-to-cycle variation in amplitude of the vocal signal.

¹⁹Harmonics-to-noise ratio (HNR) measures the proportion of periodic energy to noise in a voice, indicating voice quality.

²⁰F1, F2, and F3 are the first three formants, or frequencies of the vocal tract that make up vowel identity and timbre.

where μ_{train} and σ_{train} are computed on the training set only to prevent data leakage and $\varepsilon = 10^{-8}$ to account for near-zero variance features.

3.4 Model Architecture

The overall framework is an N-stream model, where N can be either two or three, referring to lyrical and audio modalities or lyrical, audio, and prosodic modalities. Each selected modality is encoded by its own stream, the streams interact through emotion-LSTM cells that share a single emotion vector, and the resulting candidate emotions are combined by a learned gate. While the baseline uses two streams (VGGish acoustic and ALBERT lyric), the experimental groups add a third prosodic stream, with the aforementioned variable feature sets. All streams operate on the same axis (pair or word), and each modality’s native unit is resampled to that axis.

The emotion-LSTM cell extends the standard LSTM to enable cross-modal interaction through the shared emotion vector, following the MMD-MII framework. At each time step t , a stream receives its own feature vector, the previous hidden and cell state, and the shared previous emotion vector $\mathbf{e}_{t-1} \in \mathbb{R}^{128}$, and produces a candidate emotion vector $\mathbf{e}_t^k$. The native feature dimensions differ by modality: VGGish yields $\mathbf{a}_t \in \mathbb{R}^{128}$, ALBERT yields $\mathbf{l}_t \in \mathbb{R}^{768}$, and the prosodic stream yields $\mathbf{p}_t \in \mathbb{R}^{25}$ (prosody-v1), $\mathbf{p}_t \in \mathbb{R}^{46}$ (prosody-v2), or $\mathbf{p}_t \in \mathbb{R}^{26}$ (prosody-v3).

The candidate emotions are combined by a learned gate. For two streams the gate is $g = \sigma(W[\mathbf{e}_t^m; \mathbf{e}_t^l])$, and the fused emotion is

$$\mathbf{e}_t = g \odot \mathbf{e}_t^m + (1 - g) \odot \mathbf{e}_t^l \quad (4)$$

For three streams a softmax is first performed over the streams, giving per-stream weights, and the fused emotion is

$$\mathbf{e}_t = \sum_{k=1}^N w_k \mathbf{e}_t^k \quad (5)$$

with

$$w = \text{softmax}(W[\mathbf{e}_t^1; \dots; \mathbf{e}_t^N]) \quad (6)$$

Following the MMD-MII methodology, song structure is explicitly modeled as verse and chorus structures via automatic detection, and the verse and chorus branches are combined with a learned selection (`torch.where`). The final prediction head outputs continuous valence and arousal predictions in the range $[-1, 1]$, consistent with the DEAM annotations.

3.5 Training

Models were trained on the 594 usable DEAM tracks

with mean squared error (MSE)²¹ as the loss, the AdamW optimizer²², a learning rate of 1×10^{-4} , and a batch size of 32. Training ran for at most 100 epochs²³ with early stopping on the validation loss and a patience of 10 epochs.

3.6 Evaluation

For the regression task, mean squared error (MSE) and its square root (RMSE) measure the average squared deviation between predicted and true values, with smaller values indicating more accurate predictions. The coefficient of determination (R^2) measures the proportion of variance in the ground-truth labels that the model explains, where 1 corresponds to perfect prediction and 0 corresponds to predicting only the mean. The Pearson correlation coefficient (PCC) measures the linear association between predictions and ground truth on a scale from -1 to 1 , where 1 indicates perfect linear agreement. The concordance correlation coefficient (CCC) is similar to PCC but additionally penalizes systematic offsets in scale or mean, making it a stricter measure of how closely predictions match ground-truth values.

The ablation compares one baseline against three versioned prosodic feature sets. The main comparison is the baseline (VGGish + ALBERT) against the baseline plus prosody-v2, both at pair granularity, evaluated over 25 paired folds (five partition seeds $\times$ five folds). The remaining points of comparison are exploratory and use a single seed over five folds: the baseline and prosody-v2 at word granularity, and prosody-v3 at both pair and word granularity. A further single-fold block ablation removes one prosody-v2 feature block at a time. Finally, prosody-v1 was trained under track supervision rather than the sentence supervision used by the headline arms, so its numbers are not directly comparable.

3.7 Reproducibility Notes

All random operations were seeded. The default training seed is 42. Each of the five folds within the main comparison uses a training seed of `partition_seed + fold`, and the five partition seeds are 42, 1042, 2042, 3042, and 4042, yielding 25 paired folds. The alternative versions use only partition seed 42. This maintains diversity across training environments while keeping the results reproducible through deterministic training.

²¹Mean squared error (MSE) is the standard loss function for regression, averaging the squared differences between predictions and ground-truth values.

²²The optimizer is the algorithm that adjusts the model’s weights to minimize the loss; AdamW is the most frequently used optimizer.

²³An epoch is one full pass of the training algorithm through the entire training dataset.

Table 2. Pair-level regression metrics over 25 paired folds.

Metric	Base	Base + prosody-v2
RMSE	0.1984	0.1915
R^2	0.3018	0.3465
PCC	0.5413	0.5863
CCC	0.4396	0.5076

Table 3. Pair-level treatment effect over 25 paired folds.

Metric	Delta (prosody-v2 - base)	95% CI
R^2	+0.0447	[+0.0213, +0.0681]
CCC	+0.0680	[+0.0473, +0.0887]
RMSE	-0.0069	[-0.0103, -0.0034]

4 Results

The results indicate that a treatment of prosody-v2 feature set improves every reported metric. The paired differences and their 95% confidence intervals over the same 25 folds are shown in Table Table 3.

A paired t-test gives $p = 0.0006$, indicating that these results are statistically significant. Per output, valence improves by +0.0514 in R^2 (CI [+0.0272, +0.0756]) and arousal by +0.0381 (CI [+0.0097, +0.0665]), with both improving in 22 of 25 folds.

The same treatment at word granularity, evaluated over five exploratory folds under a single partition seed, does not demonstrate a similar increase in performance. In fact, the results obtained were a +0.0213 in R^2 , with only 3 of 5 folds improving. These results indicate that word-level processing does not increase performance, at least by a statistically significant amount.

The single-fold block ablation (Figure Figure 2, panel c) removes one prosody-v2 feature block at a time at word granularity. Every removal lowers R^2 relative to the full set (0.478), with timing (-0.124), f0 (-0.110), tonality (-0.073), and loudness (-0.045) contributing most. Because this ablation uses one fold it is exploratory and thus not indicative of a guaranteed improvement, but provides a basis for future research. As a control, prosody-v1 does not improve over the baseline under track supervision (0.340 vs. 0.369 for R^2), consistent with the gain coming from the feature treatment rather than from adding a third stream.

5 Discussion

The central result of this study is a statistically supported improvement, but the improvement itself is generally modest. While this does achieve state-of-the-art performance, it does so with minor improvement to accuracy and over a different output than the MMD-MII model was designed for.

Table 4. Experiment setup and dataset counts.

Setting	Value
Labelled tracks	801
Usable tracks	594
Folds	5
Validation fraction	0.1
Test fraction	0.1
Supervision	sentence
Partition seeds	5 (main); 1 (exploratory)
Paired folds per arm	25 (main); 5 (exploratory)

Adding prosody-v2 at pair-level granularity specifically only raises R^2 by +0.0447 over the two-stream baseline, supported with a 95% confidence interval that excludes zero. Associated with this are improvements to CCC and RMSE as well. This effect is also supported by the general observation that each run generally leads to increased improvement in at least one of valence or arousal accuracy. The control of prosody-v1 shows that the gain is not explained by the mere addition of more parameters, as it uses the same three-stream architecture and similar increases in parameter count but does not improve over the baseline under track supervision.

A substantial performance divergence remains between the two prediction subtasks. It is wholly evident that arousal is predicted much more accurately than valence, as arousal R^2 is roughly 0.5, while valence R^2 is roughly 0.13-0.21. Prosody improves both outputs, but it improves valence more (+0.0514 in R^2) than arousal (+0.0381). This asymmetry is consistent with prior findings that valence is considerably more difficult to predict than arousal, because arousal tends to map more reliably onto lower-level features while valence is more reliant on the higher-level features of harmonicity, mode, lyrical content, and subtle timbral nuances[10, 23].

The pair-level gain here is an interesting piece of evidence to note. This shows that the prosodic feature treatment can contribute non-redundant signal to the cross-modal interaction on a pair-level. However, the lack of a significant increase in word-level analysis indicates that moving the interaction down to individual words may not be as useful to the task of music emotion recognition. Taken together, the two results suggest that the benefit comes from the coarser prosody-v2 feature representations rather than the finer units as initially hypothesized. This overall lends credibility to the claim that the sentence-level unit of the existing architecture is already adequate for this task.

The prediction figures (Figures Figure 3 and Figure 4) show the same pattern at 25-fold scale. Arousal predictions tend to line up to the line of identity closer than valence, with errors centered near zero and a standard deviation of about 0.19, while valence predictions are compressed

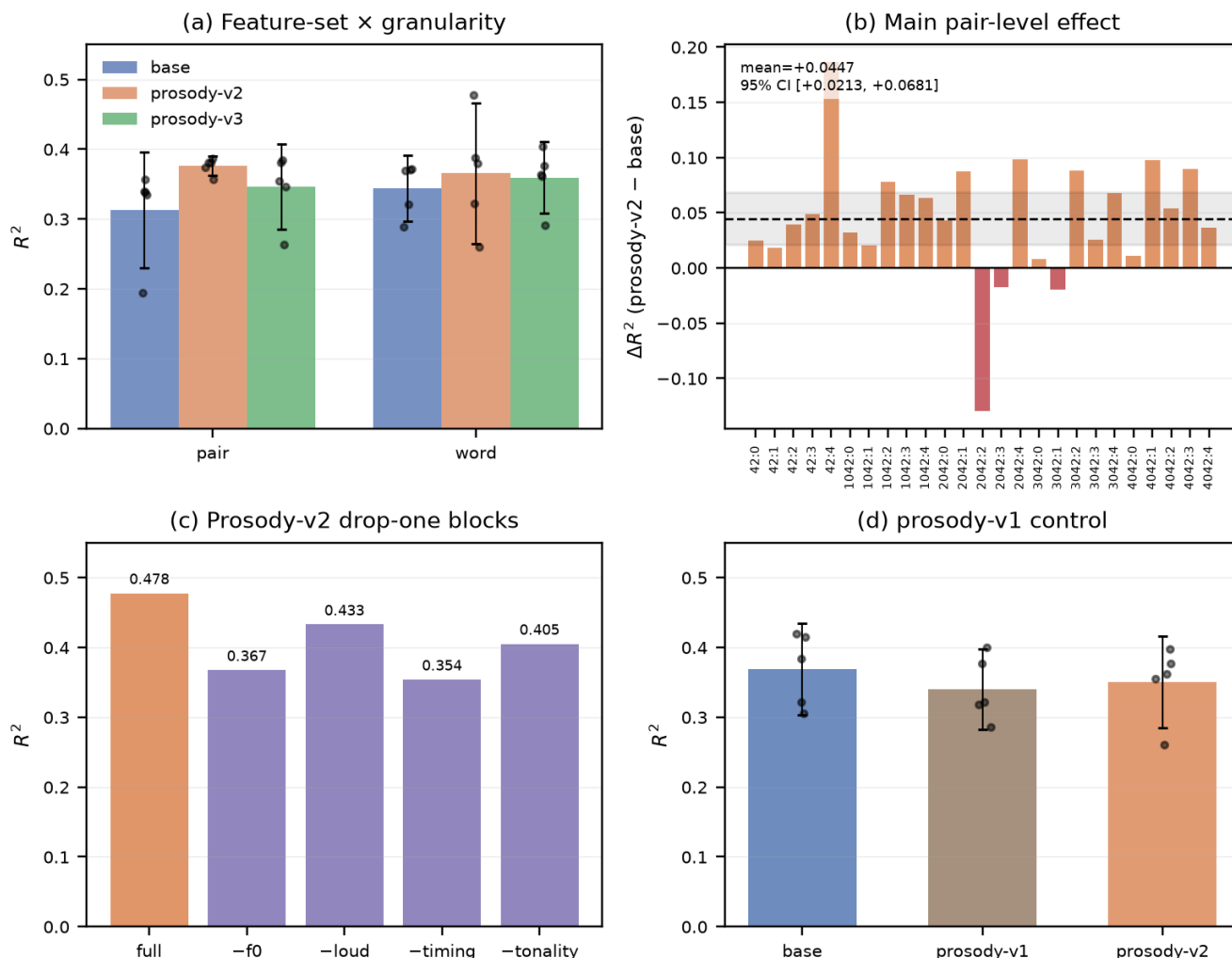


Figure 2. Ablation overview. (a) R^2 charted at pair and word granularity respectively for each set of features over five folds. Each dot is an individual fold’s measurement. (b) Per-fold R^2 difference between prosody-v2 and the baseline at pair-level granularity across the 25 paired folds. The dashed line and band are the mean and 95% confidence interval respectively. (c) Single-fold drop-one block ablation of prosody-v2 features. (d) Prosody-v1 control under track supervision.

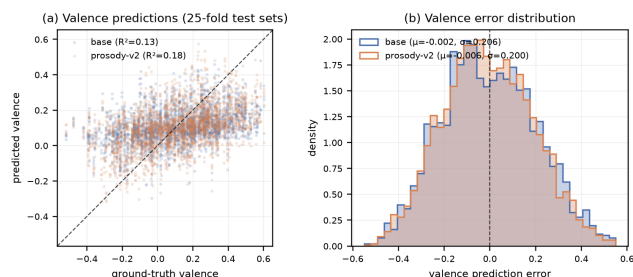


Figure 3. Valence. (a) Predicted versus ground-truth valence plotted for prosody-v2 vs. base. The dashed line is the identity. (b) Prediction-error distributions for the two runs.

toward zero. This means that tracks with a valence on the

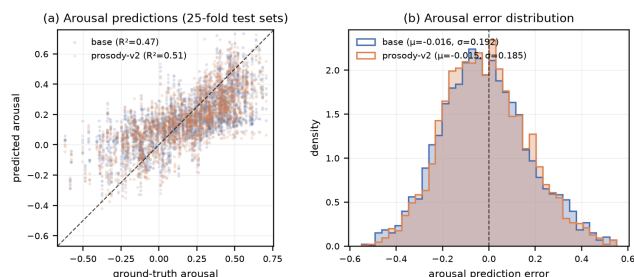


Figure 4. Arousal. (a) Predicted versus ground-truth arousal plotted for prosody-v2 vs. base. The dashed line is the identity. (b) Prediction-error distributions for the two runs.

extreme ends of the spectrum (high or low) are typically

under- and over-predicted respectively, and the error distribution of valence is generally wide relative compared to that of the arousal. However, note that adding prosody-v2 features shifts the error distributions closer to zero, with the larger relative improvement on valence.

5.1 Limitations

Several limitations qualify these results. First, the study uses a single dataset (DEAM) and a single architecture family (the N-stream emotion-LSTM), so the effect may not transfer to other corpora or models. Second, the effect itself is not very large, as the R^2 increase of 0.045 is indeed statistically supported to be significant, yet is small in absolute terms. Arousal itself still remains harder to predict than valence. Third, only the main pair-level comparison uses the full set of 25 paired folds while the word-level and prosody-v3 arms use a total of five folds. Thus, there may be a statistically significant difference between values, but the scale of performed comparison did not identify one. Finally, the dataset is small by the modern size of typical machine learning datasets, with only 594 usable tracks, which constrains generalization.

5.2 Implications

The results carry implications for MER development and for applications such as music therapy and recommendation. The pair-level gain shows that a richer prosodic feature treatment contributes a non-redundant signal to a cross-modal framework, and that the signal can be captured at the sentence-level unit already used by existing architectures rather than requiring word-level alignment. At the same time, the modest effect size, despite the statistical significance, indicates that prosodic features do not suffice for the innovation of additional feature types. In general, this improvement should be described as a step forward rather than a full solution; human performance of valence-arousal is still miles ahead of MER systems.

In the real world, the arousal-side accuracy observed here means that applications centered on detecting the energy or activation level of a track, such as music therapy session design or mood-aware recommendation, stand to benefit most. Applications that require high valence accuracy, such as automated mood tagging or affect-sensitive content moderation, will require further work, given the model’s weaker valence performance.

6 Conclusion

In this study, an N-stream framework for multimodal music emotion recognition was introduced, integrating prosodic analysis of vocals alongside existing streams of lyrical and acoustic analysis. The effectiveness of the approach was verified through valence-arousal regression under a five-fold cross-validation. Adding the prosody-v2 treatment at pair

granularity produced a modest but statistically supported improvement over the baseline of the MMD-MII architecture across 25 paired folds. At the same time, treatment at word granularity produced no reliable gain, indicating that finer granularity may not necessarily lead to improvement. Despite the limitations of dataset size, this represents a meaningful step forward in the domain of music emotion recognition.

7 Appendix

7.1 Evaluation Metric Definitions

The formal definitions of each metric referenced in Section 3.6 are provided below. In each equation, y_i denotes a ground-truth value, $\hat{y}_i$ a predicted value, $|y|$ and $|\hat{y}|$ the corresponding means, σ_y and $\sigma_{\hat{y}}$ the corresponding standard deviations, and N the number of samples.

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (7)$$

$$\text{RMSE} = \sqrt{\text{MSE}} \quad (8)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - |y|)^2} \quad (9)$$

$$r = \frac{\sum_{i=1}^N (y_i - |y|)(\hat{y}_i - |\hat{y}|)}{\sqrt{\sum_{i=1}^N (y_i - |y|)^2} \sqrt{\sum_{i=1}^N (\hat{y}_i - |\hat{y}|)^2}} \quad (10)$$

$$\rho_c = \frac{2r\sigma_y\sigma_{\hat{y}}}{\sigma_y^2 + \sigma_{\hat{y}}^2 + (\mu_y - \mu_{\hat{y}})^2} \quad (11)$$

References

- [1] Kat R. Agres, Rebecca S. Schaefer, Anja Volk, Susan van Hooren, Andre Holzapfel, Simone Dalla Bella, Meinard Müller, Martina de Witte, Dorien Herremans, Rafael Ramirez Melendez, Mark Neerincx, Sebastian Ruiz, David Meredith, Theo Dimitriadis, and Wendy L. Magee. 2021. Music, computing, and health: A roadmap for the current and future roles of music technology for health care and well-being. *Music & Science* 4 (Jan. 2021). doi:10.1177/2059204321997709
- [2] Laura-Lee Balkwill and William Forde Thompson. 1999. A cross-cultural investigation of the perception of emotion in music: Psychophysical and cultural cues. *Music Perception: An Interdisciplinary* 17, 1 (Oct. 1999), 43–64. doi:10.2307/40285811
- [3] Masataka Goto, Jin Ha Lee, and Meinard Müller. 2024. Exploring 25 years of music information retrieval: Perspectives and insights. Tutorial presented at ISMIR 2024. https://staff.aist.go.jp/m.goto/PAPER/ISMIR_2024_Tutorial.Exploring_25_Years_of_Music_Information_Retrieval.pdf Retrieved December 19, 2025.
- [4] Romain Hennequin, Anis Khelif, Felix Voituret, and Manuel Moussallam. 2020. Spleeter: a fast and efficient music source separation tool with pre-trained models. *Journal of Open Source Software* 5, 50 (2020), 2154. doi:10.21105/joss.02154
- [5] Shawn Hershey, Sourish Chaudhuri, Daniel P. W. Ellis, Jort F. Gemmeke, Aren Jansen, R. Channing Moore, Manoj Plakal, Devin Platt, Rif A. Saurous, Bryan Seybold, Malcolm Slaney, Ron J. Weiss, and Kevin W. Wilson. 2016. CNN Architectures for Large-Scale Audio

- Classification. *CoRR* abs/1609.09430 (2016). arXiv:1609.09430 <http://arxiv.org/abs/1609.09430>
- [6] Il-Young Jeong and Kyogu Lee. 2016. Learning temporal features using a deep neural network and its application to music genre classification. In *Proceedings of the International Society for Music Information Retrieval Conference (ISMIR)*. New York, NY, USA, 434–440. <https://archives.ismir.net/ismir2016/paper/000159.pdf>
 - [7] Mukkamala S. N. V. Jitendra. 2020. A review: Music feature extraction from an audio signal. *International Journal of Advanced Trends in Computer Science and Engineering* 9, 2 (April 2020), 973–980. doi:10.30534/ijatcse/2020/11922020
 - [8] Patrik N. Juslin. 2001. Communicating emotion in music performance: A review and theoretical framework. In *Music and Emotion: Theory and Research*. Oxford Academic, New York, NY, USA, 309–338. doi:10.1093/oso/9780192631886.003.0014
 - [9] Zhenzhong Lan, Mingda Chen, Sebastian Goodman, Kevin Gimpel, Piyush Sharma, and Radu Soricut. 2019. ALBERT: A Lite BERT for Self-supervised Learning of Language Representations. *CoRR* abs/1909.11942 (2019). arXiv:1909.11942 doi:10.48550/arXiv.1909.11942
 - [10] Rashini Liyanarachchi, Aditya Joshi, and Erik Meijering. 2025. A survey on multimodal music emotion recognition. arXiv:2504.18799 doi:10.48550/arXiv.2504.18799
 - [11] Beth Logan. 2000. Mel frequency cepstral coefficients for music modeling. In *Proceedings of the International Symposium on Music Information Retrieval (ISMIR)*. Plymouth, MA, USA. https://ismir2000.ismir.net/papers/logan_abs.pdf
 - [12] Barbara Pastuszek-Lipińska. 2025. The role of musical aspects of language in human cognition. *Frontiers in Psychology* 16 (March 2025). doi:10.3389/fpsyg.2025.1505694
 - [13] Jonathan Posner, James A. Russell, and Bradley S. Peterson. 2005. The circumplex model of affect: An integrative approach to affective neuroscience, cognitive development, and psychopathology. *Development and Psychopathology* 17, 3 (Sept. 2005). doi:10.1017/S0954579405050340
 - [14] James A. Russell. 1980. A circumplex model of affect. *Journal of Personality and Social Psychology* 39, 6 (1980), 1161–1178.
 - [15] Justin Salamon and Emilia Gómez. 2012. Melody extraction from polyphonic music signals using pitch contour characteristics. *IEEE Transactions on Audio, Speech, and Language Processing* 20, 6 (Aug. 2012), 1759–1770. doi:10.1109/TASL.2012.2188515
 - [16] Jaza Syed, Ivan Meresman Higgs, Ondřej Cířka, and Mark Sandler. 2025. Exploiting music source separation for automatic lyrics transcription with Whisper. In *2025 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*. 1–6. doi:10.1109/icmew68306.2025.11152264
 - [17] Robert E. Thayer. 1990. *The Biopsychology of Mood and Arousal*. Oxford Academic, New York, NY, USA. doi:10.1093/oso/9780195068276.001.0001
 - [18] William Forde Thompson, E. Glenn Schellenberg, and Gabriela Husain. 2001. Arousal, mood, and the Mozart effect. *Psychological Science* 12, 3 (May 2001), 248–251. doi:10.1111/1467-9280.00345
 - [19] Klára Vicsi and György Szaszák. 2010. Using prosody to improve automatic speech recognition. *Speech Communication* 52, 5 (May 2010), 413–426. doi:10.1016/j.specom.2010.01.003
 - [20] Michael Wagner and Duane G. Watson. 2010. Experimental and theoretical advances in prosody: A review. *Language and Cognitive Processes* 25, 7–9 (May 2010), 905–945. doi:10.1080/01690961003589492
 - [21] Jingyi Wang, Alireza Sharifi, Thippa Reddy Gadekallu, and Achyut Shankar. 2024. MMD-MII model: A multilayered analysis and multimodal integration interaction approach revolutionizing music emotion classification. *International Journal of Computational Intelligence Systems* 17, 1 (April 2024). doi:10.1007/s44196-024-00489-6
 - [22] Liang Yang, Zhexu Shen, Jingjie Zeng, Xi Luo, and Hongfei Lin. 2023. COSMIC: Music emotion recognition combining structure analysis and modal interaction. *Multimedia Tools and Applications* 83, 5 (July 2023), 12519–12534. doi:10.1007/s11042-023-15376-z
 - [23] Yi-Hsuan Yang and Homer H. Chen. 2012. Machine recognition of music emotion: A review. *ACM Transactions on Intelligent Systems and Technology* 3, 3 (May 2012), 1–30. doi:10.1145/2168752.2168754
 - [24] Marcel Zentner, Didier Grandjean, and Klaus R. Scherer. 2008. Emotions evoked by the sound of music: Characterization, classification, and measurement. *Emotion* 8, 4 (2008), 494–521. doi:10.1037/1528-3542.8.4.494